engine coolant temperature messages to the instrument cluster. For further diagnosis of the engine coolant temperature gauge or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the check gauges indicator due to a high or critical engine temperature gauge reading, it may indicate that the engine or the engine cooling system requires service. For proper diagnosis of the engine coolant temperature sensor, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the engine coolant temperature gauge, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

FOG LAMP INDICATOR

DESCRIPTION

Fig. 13 Fog Lamp Indicator

A fog lamp indicator is standard equipment on all instrument clusters, but is only functional on vehicles equipped with the optional front and/or rear fog lamps (Fig. 13). The fog lamp indicator is located near the upper edge of the instrument cluster, to the left of center. The fog lamp indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Fog Lamps” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A green Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in green through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The fog lamp indicator is serviced as a unit with the instrument cluster.

OPERATION

The fog lamp indicator gives an indication to the vehicle operator whenever the optional front and/or rear fog lamps are illuminated. The availability of the front fog lamps, or rear fog lamps options varies by the market for which the vehicle is manufactured. This indicator is controlled by a hard wired input to the cluster from the fog lamp switch circuitry of the left multi-function switch. The fog lamp indicator Light Emitting Diode (LED) is grounded on the instrument cluster electronic circuit board at all times. Therefore, the LED will be on any time the front or rear fog lamps are illuminated, regardless of the ignition switch position. The LED only illuminates when it is provided battery current by the fog lamp switch circuitry of the left multi-function switch.

The fog lamp switch circuitry of the left multi-function switch is connected in series between a fused B(+) fuse in the Power Distribution Center (PDC) and the front or rear fog lamp feed input to the instrument cluster through the fog lamp relay, which is also in the PDC. The fog lamp switch input to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. For proper diagnosis of the fog lamp switch and circuits, (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/ LEFT MULTI-FUNCTION SWITCH - DIAGNOSIS AND TESTING).

DIAGNOSIS AND TESTING - FRONT / REAR FOG LAMP INDICATOR

The diagnosis found here addresses an inoperative front/rear fog lamp indicator condition. Before beginning this test, confirm the functionality of the front or rear fog lamp system. If no fog lamp system problem is found, the following procedure will help to locate an open in the front or rear fog lamp feed circuit between the fog lamp relay and the instrument cluster. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C1) for the instrument cluster from the connector receptacle on the back of the cluster housing.

(2) Reconnect the battery negative cable. Turn the fog lamps on by pulling out the control knob on the end of the left-multi-function switch control stalk. Check for battery voltage at the front/rear fog lamp